3.

Axis	Vertex	Focus	Equation of directrix	Length of latus rectum
$y = 4$	$V(1, 4)$	$F(3, 4)$	$x + 1 = 0$	8 units

4.

Problem	Axis	Vertex	Focus	Equation of directrix	Length of latusrectum
(a) $y^2 = 20x$	$y = 0$	$V(0, 0)$	$F(5, 0)$	$x = -5$	20 units
(b) $x^2 = 8y$	$x = 0$	$V(0, 0)$	$F(0, 2)$	$y = -2$	8 units
(c) $x^2 = -16y$	$x = 0$	$V(0, 0)$	$F(0, -4)$	$y = 4$	16 units

5. $(x - 15)^2 = 5(y - 55)$. The output and the average cost at the vertex are 15 kg. and ₹ 55.

6. when $x = 5$ months

Exercise 3.7

1	2	3	4	5	6	7	8	9	10	11	12	13
(b)	(c)	(c)	(c)	(b)	(c)	(a)	(c)	(a)	(b)	(c)	(a)	(d)
14	15	16	17	18	19	20	21	22	23	24	25	
(a)	(a)	(d)	(a)	(c)	(b)	(b)	(a)	(d)	(b)	(b)	(b)	

Miscellaneous problems

1. $3x + y - 8 = 0$ 2. $y = 6x + 3000$ 4. $p = 1, p = 2$ 6. $a = 9, b = 8$

7. $(-1, -2)$ is on the line, $(1, 0)$ is above the line and $(-3, -4)$ is below the line

8. $(-2, -7)$ 9. $y^2 = -\frac{9}{2}x$

10. Axis : $y = 2$, Vertex : $(1, 2)$, Focus : $(2, 2)$,
Equation of directrix : $x = 0$ and Length of latus rectum : 4 units

4. TRIGONOMETRY

Exercise 4.1

1.(i) $\frac{\pi}{3}$ (ii) $\frac{5\pi}{6}$ (iii) $\frac{4\pi}{3}$ (iv) $\frac{-16\pi}{9}$

2.(i) $22°30'$ (ii) 324^0 (iii) $-171°48'$ (iv) $110°$

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